Log-Space Importance Model for Ethereum

Deep Funding GG24 · Level I · by Anas

Aura

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github.com/i-anasop/L3

98	REPOS WEIGHTED
48.6%	JURY VARIANCE EXPLAINED
log	NATIVE TARGET SPACE
LOO	HONEST VALIDATION

> Aura assigns relative importance weights to the 98 repositories Ethereum depends on. It is built by **reading the contest's scoring code** rather than guessing at what the jury rewards — and that single decision shaped every modeling choice that follows.

1 The problem, and why it rewards generalization

Assign 98 repos a weight summing to 1. The ground truth is a human jury answering *pairwise* questions — "is A more important than B, and by how much?" New jury data arrives during the contest: a portion updates the live leaderboard, and **the rest is held out for final scoring.** Final placement therefore rewards **generalization, not memorization** — which is why reproducing the public weights to many decimals (≈0 on the public board) collapses on the board that pays.

2 Reading the scoring

```
def cost_function(logits, samples):
    return sum((logits[b] - logits[a] - c) ** 2 for a, b, c in samples)
```

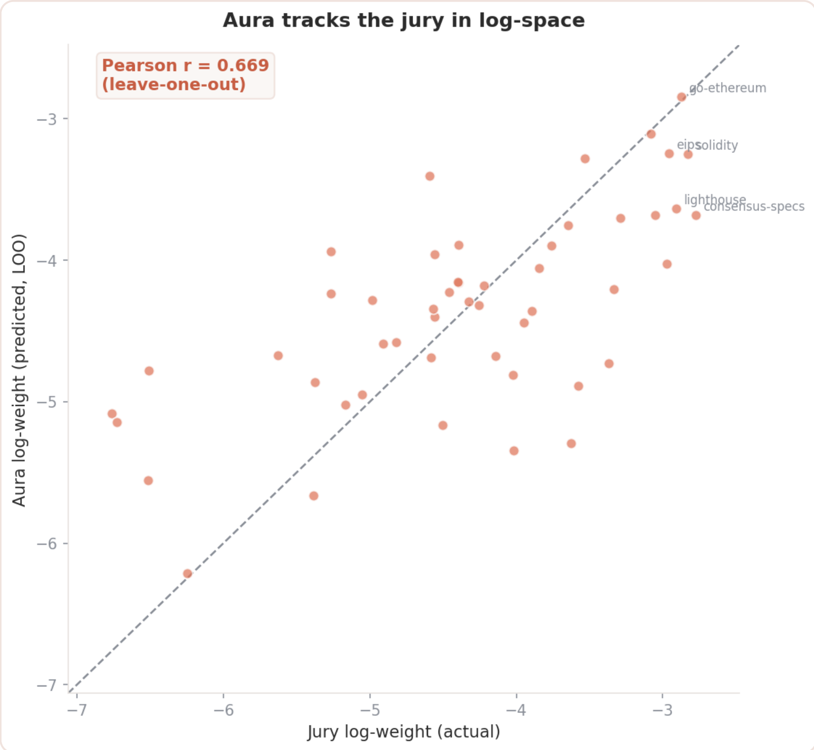
Each sample `(a,b,c)` gives two repos and `c`, the **log** of how much more important `b` is than `a`. With $x_i = \log w_i$, the jury's own weights minimize:

$$J(x) = \Sigma_{(a,b,c)} \left(x_b - x_a - c \right)^2$$

This least-squares problem on a difference operator has a one-dimensional null space — adding a constant to every x_i leaves `J` unchanged. Three consequences define Aura: **(1)** the target is log-weight; **(2)** the score is **scale-invariant** — only the shape matters, normalization is irrelevant; **(3)** the **spread** of log-weights is the single highest-leverage parameter.

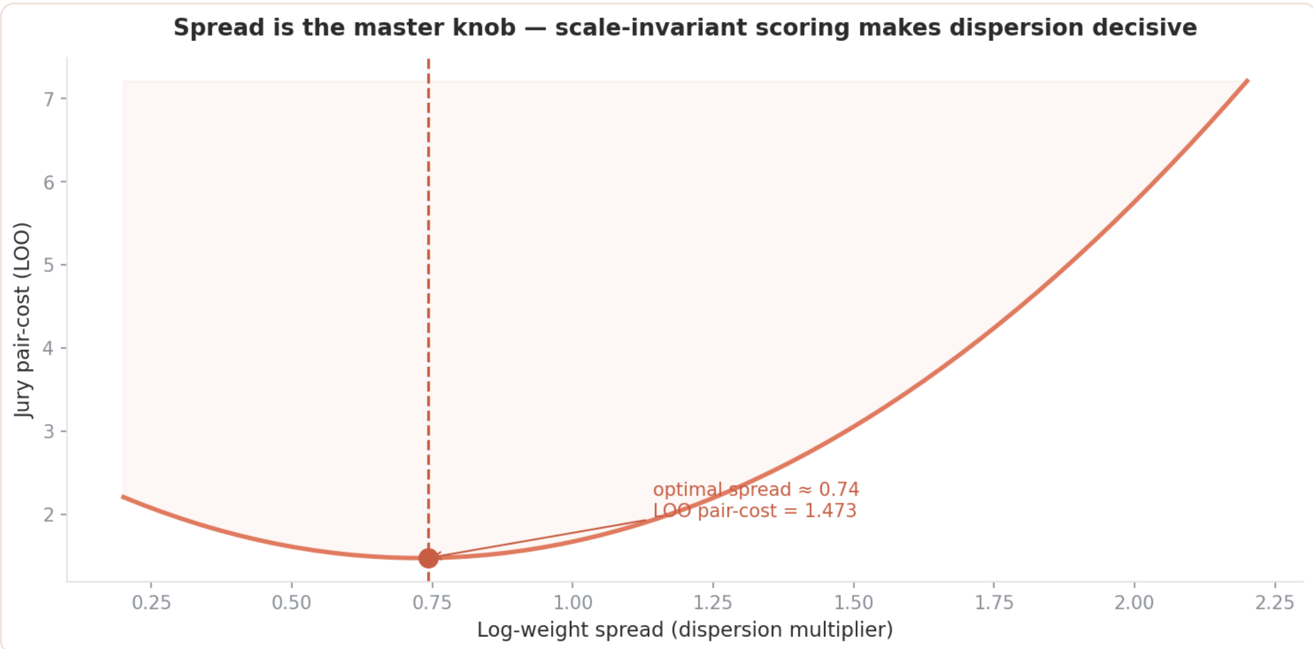
3 The model

A regularized **log-space ridge regression** trained on the 50 repos with public jury weights, predicting all 98.




```
features → standardize → ridge (α=10) → exp() → calibrate log-spread → normalize
```

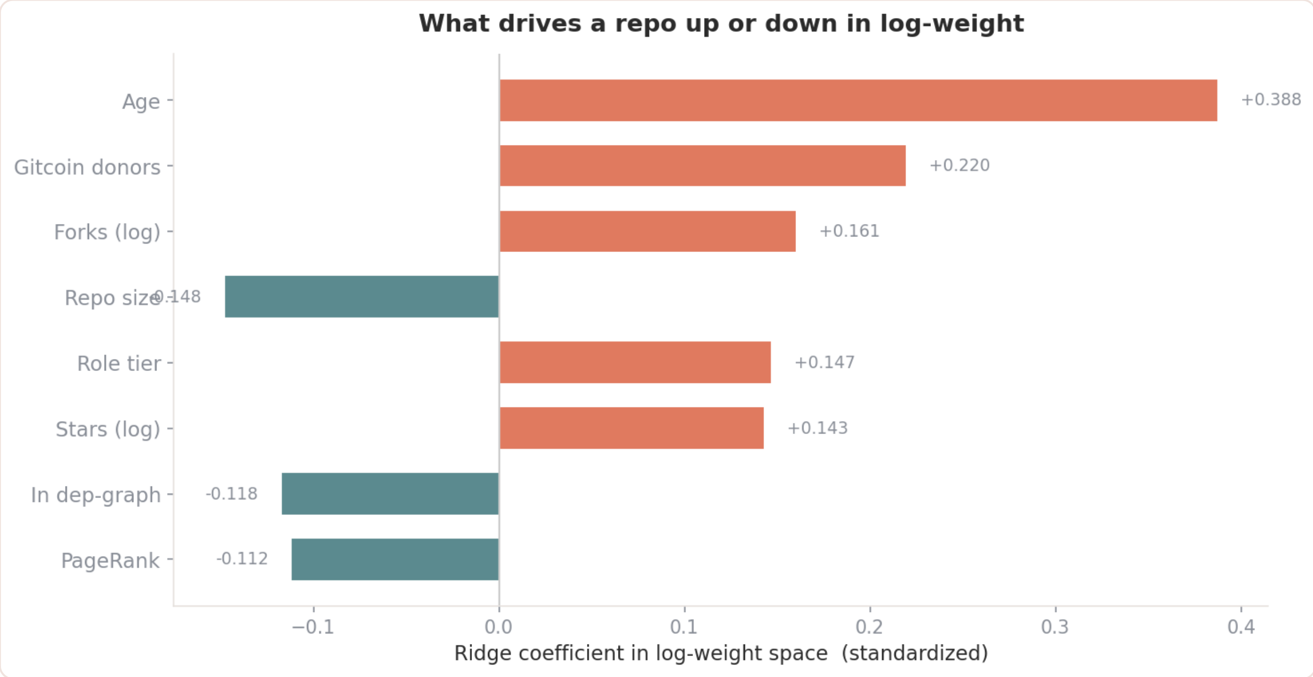
Two deliberate choices: **heavy L2** (only 50 labels — a gradient-boosted ensemble was tested and *lost* to ridge in CV), and **spread calibration** tuned directly against held-out jury cost.



Under-dispersed weights leave the jury's strong preferences unexpressed; over-dispersed weights overshoot. The minimum sits at a spread of **0.74**.

4 Features

Eight features, chosen by forward selection against the real jury cost.

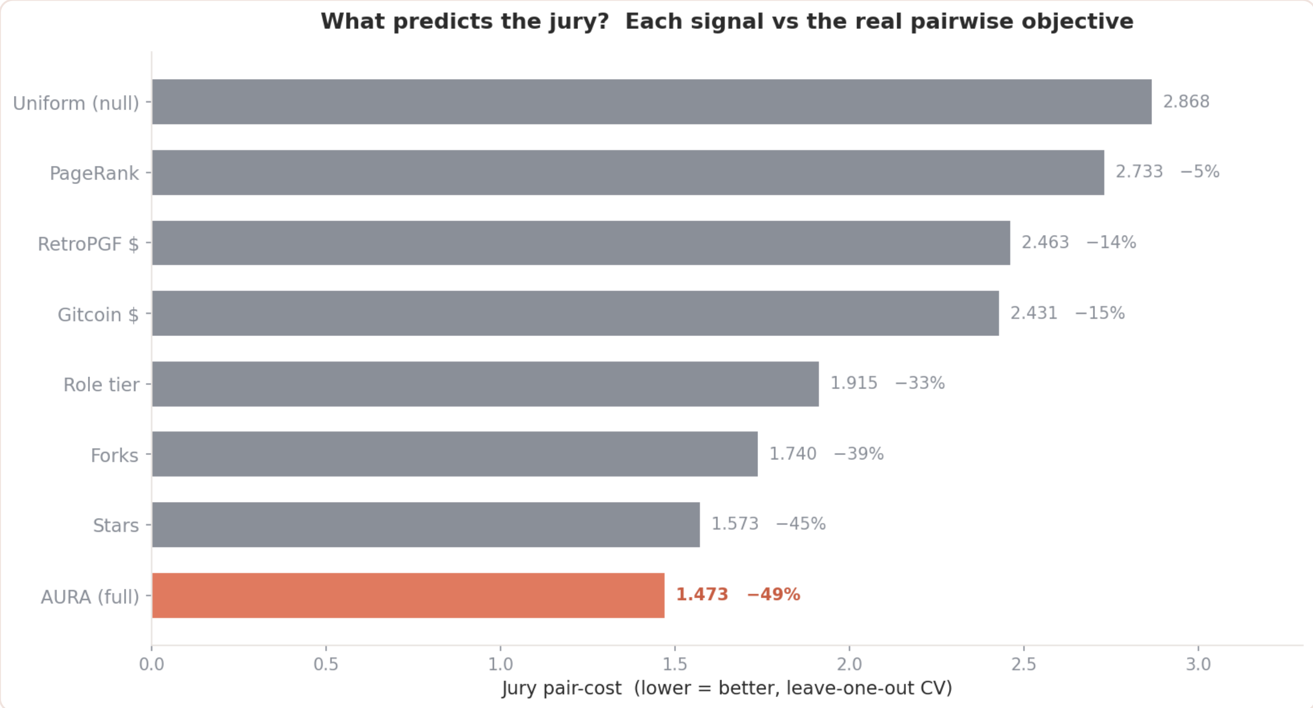


Reading this: these are *partial* coefficients in the joint model. Adoption features (stars/forks/size/age) are correlated, so ridge spreads the shared signal — `age` leads while `size` / `pagerank` go mildly negative after the others absorb adoption. This is conditional effect, not standalone power (for which stars leads, §5).

Feature	Source	Signal
log_stars, log_forks, log_size	GitHub (live)	adoption & code surface area
age_days	GitHub (live)	project maturity
tier_prior	hand-coded roles	ecosystem function
pagerank, has_graph	dependency graph	structural centrality
gitcoin_donors	OSO funding	community funding signal

5 Validation against the real jury

Every model scored on the **171 public pairwise jury comparisons** with the exact cost, under **leave-one-out CV** — the post-event generalization regime.



Model	Jury pair-cost	vs null
Uniform (null)	2.868	—
PageRank only	2.733	−5%
RetroPGF $ only	2.463	−14%
Gitcoin $ only	2.431	−15%
Role tier only	1.915	−33%
Forks only	1.740	−39%
Stars only	1.573	−45%
AURA (full)	**1.473**	**−49%**

Aura explains **48.6%** of jury disagreement and beats every individual signal. An unconstrained least-squares fit directly on the 171 comparisons (59 free params, overfit) reaches 0.55 — the in-sample ceiling no generalizing model can touch.

6 What the model learned, honestly

- **Adoption dominates.** `log_stars` alone reaches −45%; the jury tracks real usage more than any other single feature.
- **Funding helps, modestly.** RetroPGF/Gitcoin are genuine signals (−14/−15%) but sparse across the 98 repos and outranked by adoption.
- **Graph centrality is weak here.** PageRank alone barely beats null (−5%) — the public dep-graph densely covers only ~34 of 98 repos. It helps at the margin, not as the engine.
- **Simpler generalizes better.** Gradient boosting lost to ridge in LOO-CV (1.56 vs 1.47). With 50 points, regularization beats capacity.
- **Spread calibration is real.** Tuning dispersion moves pair-cost by >0.5 — the highest-leverage knob, a direct consequence of scale-invariant scoring.

7 Predicted importance